

1) d) minimize $Z = x_1 - 3x_2 - 2x_3$

Subject to

$$3x_1 - x_2 + 2x_3 \leq 7$$

$$2x_1 - 4x_2 \geq 12$$

$$-4x_1 + 3x_2 + 8x_3 = 10$$

$x_1, x_2 \geq 0$ x_3 unrestricted in sign.

Sol:

Let y_1, y_2 and y_3 be the dual variables corresponding to three primal constraints in the given order.

① Since the given LPP is of minimization all the constraints should be converted to $\geq$ type

② Also, here the third constraint of the primal is an equation, the third dual variable will be unrestricted in sign (ie y_3 is unrestricted).

The dual of the given LPP can be formulated as

Maximize $Z_y = -7y_1 + 12y_2 + 10y_3$

subject to

$$-3y_1 + 2y_2 - 4y_3 \leq 1$$

$$y_1 - 4y_2 + 3y_3 \leq -3$$

$$-2y_1 + 8y_3 \leq -2$$

and $y_1, y_2 \geq 0$;

y_3 unrestricted in sign.

2) Write the dual of the following LPP (Primal) and find the optimal solution of the dual by solving the primal LPP.

$$\text{Maximize } Z = 4x_1 + 3x_2$$

$$\begin{aligned} \text{Subject to } & x_1 + x_2 \leq 8 \\ & 2x_1 + x_2 \leq 10 \\ & x_1, x_2 \geq 0. \end{aligned}$$

Sol:

$$\text{Max } Z = 4x_1 + 3x_2 + 0s_1 + 0s_2$$

$$\begin{aligned} \text{Subject to: } & x_1 + x_2 + s_1 = 8 \\ & 2x_1 + x_2 + s_2 = 10 \\ & x_1, x_2, s_1, s_2 \geq 0. \end{aligned}$$

Iteration: 1

Basis	C_B	4	3	0	0	R.H.S	min ratio
		x_1	x_2	s_1	s_2		
s_1	0	1	1	1	0	8	8
s_2	0	2	1	0	1	10	5
		↑					
$Z_j - C_j$		-4	-3	0	0		

Iter-2:

Basis	C_B	4 x_1	3 x_2	0 s_1	0 s_2	R.H.S	min ratio
s_1	0	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	3	6
x_1	4	1	$\frac{1}{2}$	0	$\frac{1}{2}$	5	10
$Z_j - C_j$		0	-1	0	2		

Iter-3:

Basis	C_B	4 x_1	3 x_2	0 s_1	0 s_2	R.H.S	min ratio
x_2	3	0	1	2	-1	6	-
x_1	4	1	0	-1	1	2	-
$Z_j - C_j$		0	0	2	1		

$Z_j - C_j \geq 0 \quad \forall j \Rightarrow$ optimality obtained

$$x_1 = 2 + x_2 = 6 \Rightarrow Z_{\max} = 26$$

Dual of the given problem:

$$\text{Min } W = 8y_1 + 10y_2$$

$$y_1 + 2y_2 \geq 4$$

$$y_1 + y_2 \geq 3$$

$$y_1, y_2 \geq 0.$$

now the optimal dual variables
are obtained using

$$\bar{y} = C_B \cdot B^{-1}$$

where $C_B \Rightarrow$ Co-efficient of basic
variables

$$\text{ie) } C_B = [3 \ 4] \quad \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

and $B^{-1} \Rightarrow$ the columns under
the slack variable

$$\text{ie) } B^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\Rightarrow y = [2 \ 1]$$

$$\Rightarrow y_1 = 2 \quad \& \quad y_2 = 1$$

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$$\text{Min } W = 26$$